1. Software Project Management.

* Management is an essential activity for the Computer based systems and provide product.

* It includes Planning, monitoring, Control of People, Process and various events that occur during the development of software.

* The Project management is carried out with the help of 4P's.

⇒ People
⇒ Product
⇒ Process
⇒ Project.

Management spectrum.

The successful Project management is done with the help of these four factors where the help of these four factors where the order of these elements is not arbitary. Project manager has to motivate the communication between stakeholders. He should also prepare a Project Plan for the success of the Product.

The People.

* People factor is important in software industry.

The Software Engineering Institute (SEI) has developed the People management capability maturity model (pm-cmm)

→ Recruitment
→ Selection
→ Performance Management.
→ Training Compensation
→ Career development.

The Product.

Before Planning:

1. Product objectives and scope must be established.
2. Alternative solution should be considered.
3. Management Constraints must be identified.

* Developer and customer must communicate with each other in order to define the objectives and scope of Product

* Done as the first step in requirement gathering and analysis.

The Process:

* The software process provides the framework from which the software development plan can be established.

* Different task sets, milestones, work products, and quality assurance points enable framework activities to adapt the software requirements and certain characteristics of software project

* SQA and SCM are conducted.

The Project

* To manage the complexity of the project, we conduct planned and controlled software projects.

* In order to successfully manage a software project, we must understand what can go wrong and how to do it right.

* There are some indicators that imply poor management of the projects.

2. COCOMO I Model.

⇒ It is the most widely used software estimation models in the world.

⇒ Developed in 1981 by Barry Boehm to give an estimate of the number of man-months it will take to develop a software product.

⇒ COCOMO stands for constructive Cost Model.

It has three different models.

* Basic model.
* Intermediate model.
* Detailed model.

It has three Classes:

* Organic mode.
* Embedded Projects.
* Semi-detached Project.

Basic Model:

It uses lines of code.

$E = a_b (KLOC)^{b_b}$

$D = c_b (E)^{d_b}$

$P = E/D$.

E - Effort applied

D - development time in chronological weeks KLOC - Kilo line of code.

P - Total number of Person.

Intermediate Model.

The formula for effort calculation can be

$E = a_i (KLOC)^{b_i} \cdot EAF$ Person months.

Software Project	a_i	b_i.
Organic	3.2	1.05
Semi detached	3.0	1.12
Embedded	2.8	1.20.

Detailed Model:

1. Requirements Planning and production Design (RPD)

2. Detailed Design (DD)

3. Code and unit Test (CUT)

4. Integrate and Test (IT).

Cocomo II Model.

COCOMO-II is applied for modern software development. Practics addressed for the Project in 1990's and 2000's.

1. Application Composition Model.

$$DM = (NAP (1 - \%.reuse/100)) / PROD$$

where.

PM means effort required in terms of Person method.

NAP means no of application Points required.

%. reuse means reused components in the project.

PROD is the object point productivity.

2. An early design model.

$$Effort = A \times size^{B} \times M$$

where

$$A = 2.94$$

$$PM = 2.94 \times size^{B} \times M$$

$$M = RUSE \times PDIF \times PERS \times DREX \times SCED \times FILL.$$

3. Reuse Model.

The effort required for automatically generated code is

PM = (ASLOC x AT/100) ATPRO

where

AT is the Percentage of generated code.

ESLOC : ASLOC x (1-AT/100) x AAM

ESLOC - Equivalent no of lines of new source.

ASLOC - source line of code.

AAM - Adaption Adjustement Multiplier.

4. Post Architecture Model.

Effort : A x Size^B x M

Components.

1. The estimate about new lines of code that is added in the program.
2. Equivalent number of source lines of code used on reuse model
3. The estimate amount of code being modified.

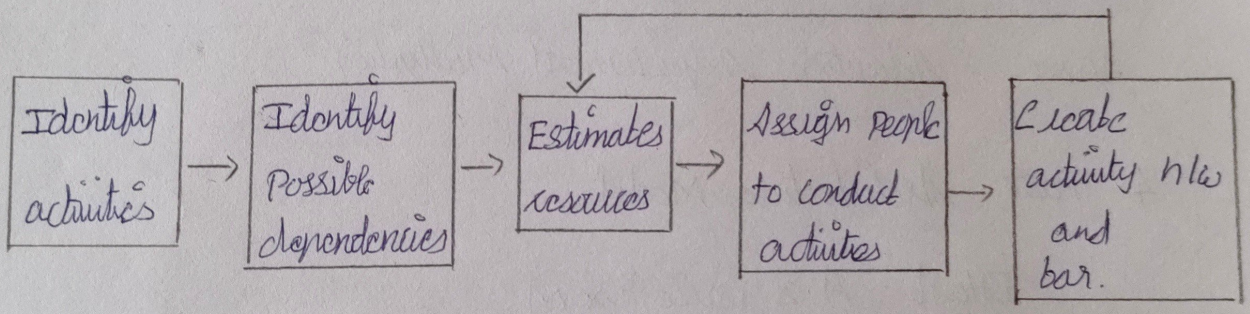
3. Project scheduling.

* The manager has to estimate the time and resource of the project.

* All project must be arranged in coherent sequence.

* During project scheduling, the total work is seperated into various small activities.

* For efficient performance some activities are conducted in parallel.



various resources are:

i) Human Effort.

ii) sufficient disk space.

iii) specialized hardware.

iv) software technology.

v) Travel allowance required by the project staff.

The Project development stage are:

i) People may leave or remain absent.

ii) Hardware may get failed.

iii) Software resource may not be available.

Relationships between people and Effort.

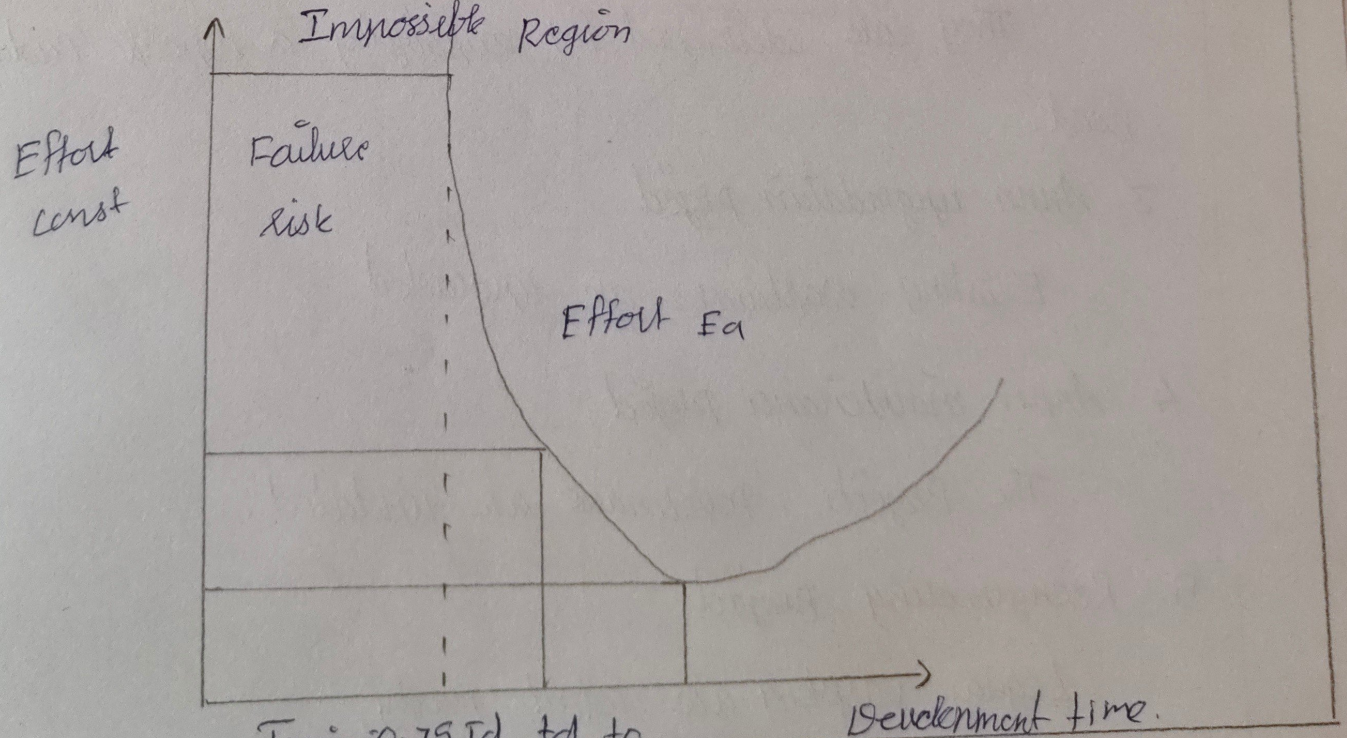
Putnam first studied the on how much staffing is required for the projects. He extended the work of worden who had earlier investigated the staffing pattern.

$$Ea = m(td^4 / td^4).$$

Ed - Effort in person months.

td - normal delivery time.

ta - actual delivery time.



Task sets:

It is a collection of software engg work task, milestones and work products that must be accomplished to complete particular project.

There is no single task that is appropriate for all the project but for developing large, complex project the set of tasks are required.

Various types of projects are.

1. Concept development project.

In this new business ideas or the apps based on new tech are to be developed.

2. New Appn development project.

They are developed for satisfying a special customer need.

3. Appn upgradation project.

Existing softwares are upgraded.

4. Appn maintenance project.

The Projects, Performance are updated.

5. Reengineering project.

Legacy system are rebuit Partly.

Task network.

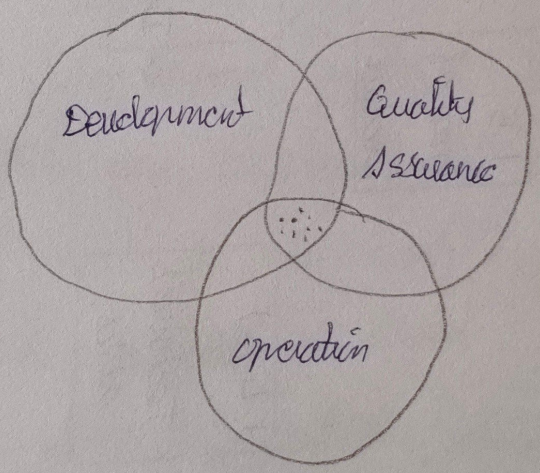
Small unit of work

contains nodes and product development.

4 i) DevOps.

DevOps is a practice in which development and operation engineers participate together in the entire life cycle activities of system development from design and implementation to product support.

The term DevOps is derived from "Software Development" and information technology operations.



DevOps.

DevOps is a set of processes and methods from three Department.

IT operation

Development

Quality Assurance.

1. Operations:

Seeks organizational stability.

2. Developers:

Seeks change.

3. Testers:

Seeks risk reduction.

Adoption of Devops is derived by various factor are demanded for an increased rate of Production releases - from application and business unit Stakeholders.

Motivation:

The Goal of Devops are as follows:

1. To make complex process increasingly programmable and dynamic.
2. Fast reliability of the product.
3. Lower failure rate of new release.
4. Shortend lead time between fixes.
5. To reduce work on program.

Benefits:

Various benefits of Devops are

Technical Benefits

Cultural Benefits.

Business Benefits.

Agility and Devops:

Basically Agile and Devops are similar but there lies some difference.

Thus in Devops there are an increased number of releases.

This actually brings the continuous delivery approach.

Deployment pipeline.

A pipeline is a set of automated process that allows developers and Devops professionals to reliably and efficiently compile, build and deploy their code to their production computing platform.

Various components of a Pipelines are.

1) Build automation.
2) Test automation.
3) Deploy automation.

There are two important Concepts.

1) Continious Integration (CI)
2) Continious Delivery (CD).

1) Continious Integration.

- It is the practice of automatically the integration of code change from micro-scope developers or testers into a single software projects.

2. Continous delivery.

- Continous delivery means a developers change to an application are automatically bug tested and uploaded to a repository.

- Deployment pipelines are logically designed was.

Overall architecture

* DevOps is a practise of operations and dev development.

* There are different phases of DevOps architecture

* They are namely Plan, code, build, test, Integrate deploy, operate, monitor.

DevOps Lifecycle.

1) Continuous development

2) continuous integration

3) Continuous testing

4) Continuous monitoring

5) continuous feedback

6) continuous deployment.

4 ii) Cloud as a platform.

Cloud as a platform is a concept that refers to using cloud computing infrastructure and services as a function for developing, deploying and running application and services.

Benefits of using cloud as a platform.

1) Flexibility
2) Cost Effective
3) Scalable and Elastic
4) Secure
5) High performance.

There are the five types of Benefits of using cloud as platform.

Operations:

It provides, different services of the cloud such as Infrastructure as a service (Iaas), Platform as a service (Paas), software as a service (saas).

Iaas:

It means delivering Computing Infrastructure on demand.

Example: Aws and Microsoft Azure.

Paas:

It means a service where a third party provides provides both hardware and software tools to the client.

Example: Facebook.

Saas:

It is the model that hosts software to make it available to clients. It is used by the end user.

Example of cloud service: Google Apps.